

Introducing Augmented Post-it: An AR Prototype for Engaging Body Movements in Online GPT-Supported Brainstorming

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ABSTRACT

Recent web-based brainstorming tools utilize pre-trained language models to support ideation. However, body movements such as walking and arm motion, which are known to enhance creativity, are generally limited in web-based brainstorming. Thus, we propose “Augmented Post-it,” a novel augmented reality (AR) interaction system that enables body movements while brainstorming using a generative pre-trained transformer (GPT). With Augmented Post-it, ideas uttered by users are structured via GPT and extended into a spatial representation that encourages divergent thinking. This study contributes to future GPT-based AR brainstorming by extending the thinking ability of users in ways that engage their body movements to enhance creativity.

CCS CONCEPTS

• Human-centered computing; • Mixed / augmented reality.;

KEYWORDS

Augmented reality, Brainstorming, Creativity, GPT

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1 INTRODUCTION

Pre-trained language models (PLMs) are trained on large textual datasets collected from various sources including books and web datasets. Recently, brainstorming tools have been developed to support creative ideation using PLMs [1]. Such tools understand the texts written by users and accordingly perform various interactions such as generating ideas [5] and displaying related images

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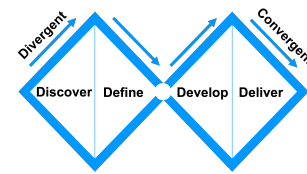


Figure 1: Design Council's Double Diamond [9]. The two diamond shapes represent the process of exploring a problem more broadly or deeply (divergent thinking) and then taking focused action (convergent thinking).

[2–4]. However, the effectiveness of online brainstorming is limited because users do not generally engage in bodily movements during the activities. Nonetheless, previous research suggests that body movements, including walking and arm motion, can enhance creativity [6–8]. Hence, tools that integrate body movements into artificial intelligence (AI)-based brainstorming are required. Thus, we propose a novel augmented reality (AR) interaction system, “Augmented Post-it,” to engage body movements in generative pre-trained transformer (GPT)-based brainstorming. In this poster, we develop multiple AR prototypes as the first step to facilitate divergent thinking and propose a novel system that integrates GPT-based brainstorming with AR experiences to support user creativity in a holistic manner.

2 DESIGN PROCESS

When a double-diamond model is used for design thinking, divergence plays a vital role in discovering and creating novel ideas by uncovering latent assumptions and offering different possibilities for problem-solving (Figure 1). However, specific cultural contexts affect the process of divergent thinking [11]. Generally, people in East Asian cultures, such as Japan, do not strictly separate thinking from feeling as compared with European–American culture but instead emphasize the role of embodied ways of thinking, feeling, and acting [12]. Thus, divergent-thinking interfaces that enable individuals to proactively engage in bodily movements and sensory-inductive experiences are likely to resonate with users in East Asian cultures.

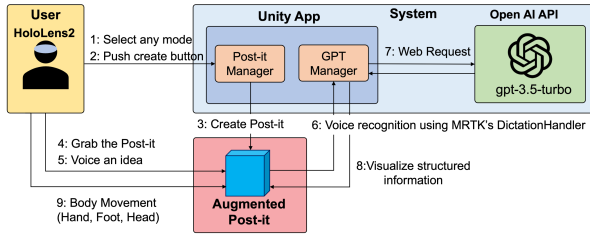


Figure 2: System overview. Voice information is converted into text and sent to the gpt-3.5-turbo model along with prompts. To access the information visualized in the virtual space, the user must engage in bodily movements.

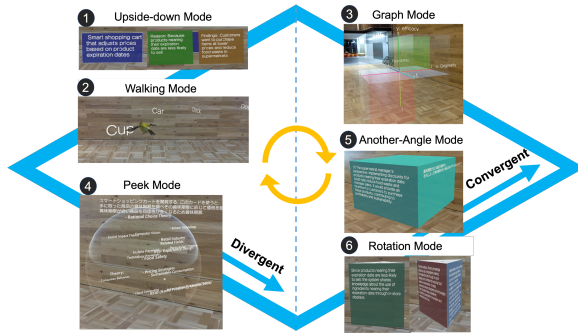


Figure 3: Proposed Augmented Post-it supports divergent thinking in GPT-based brainstorming. Repeating these modes leads to more creative ideation.

3 SYSTEM PROTOTYPE

3.1 System Overview

The AR prototype presented in this poster is developed using Microsoft HoloLens2, Unity 2022.3.4f1 LTS, and MRTK2-Unity. Figure 2 presents an overview of the proposed system. While brainstorming, the first trigger for using the Augmented Post-it is established when a user selects any mode and grabs a digital Post-it. Upon grabbing the Post-it, the ideas voiced by the user are recorded, and the voice information is converted into texts through voice recognition. Thereafter, the converted text is embedded in a predefined prompt and sent to the gpt-3.5-turbo model of the Open AI API. Structured information is visualized in a virtual space, and the user can access the augmented information by any body movement. Figure 3 shows the different developed modes, which have been discussed as follows.

3.2 Mode 1: Upside-down Mode

This mode enables hand movements of users and is inspired by conceptual metaphor theory [10], which states that abstract concepts can be understood by comparing them to concrete concepts. For example, the metaphor “seeing the background of an idea” is understood by connecting the background of an idea (reasons, needs) to the spatial backside of the idea. In this mode, the ideas of the users appear on the front, similar to a typical Post-it, while the back of the post displays the reasons for the ideas and related essential

needs. Users can access the information by flipping the Augmented Post-it using their hands.

3.3 Mode 2: Walking Mode

This mode enables users to walk while brainstorming. In this mode, a small bird appears in front of the users. By following the bird, the users can discover words that are randomly generated by the GPT. Thereafter, the users generate new “out-of-the-box” ideas by establishing connections between the brainstorming theme and unrelated words. This problem-solving technique is based on Random Input, which was invented by Dr. Edward de Bono. It is a method of brainstorming by randomly selecting nouns from a dictionary and using them as a starting point, allowing participants to step outside of their usual thinking patterns and come up with new solutions to a problem.

3.4 Mode 3: Graph Mode

This mode encourages walking and head movements by a user. The ideas the user utters are automatically evaluated on three scales and visualized on a graph on three dimensions. The user can observe them while walking around as if in a museum. Figure 3 (mode 3) shows the three scales of originality, efficacy, and feasibility, and the ideas are rated on a scale of -1~1.

3.5 Mode 4: Peek Mode

This mode encourages walking and hand movements. Idea-related (themes, theories, and technical keywords) information is displayed inside a virtual bubble object and cannot be observed from the outside. To access the information, users must hold their hand over the bubble, promoting movement.

3.6 Mode 5: Another-Angle Mode

This mode encourages perspective-taking. Generally, people view things from well-learned perspectives and are prone to bias. In this mode, three different perspectives on an idea are attributed to a teacher, an elderly person, and a psychologist or another person. These perspectives are displayed on the side of the Post-it in the form of a cube. Hence, users have to visit that side to learn about different perspectives.

3.7 Mode 6: Rotation Mode

This mode encourages hand movements in users. Positive and negative feedback and ideas are displayed on each side of a triangular pillar. Users have to hand pinch to rotate the triangular pillar-shaped Post-it to access the information.

4 CONCLUSION & FUTURE WORK

This poster presents our ongoing prototype of “Augmented Post-it,” a novel brainstorming experience that combines GPT and AR with several implementable features that encourage body movements in users during brainstorming. As these interfaces are still in the prototyping stage, extensive user testing and experimentation are required to assess their impact on creativity. Additionally, we plan to investigate cultural differences in the use of the Augmented Post-it, which enables body movements during brainstorming for creative activities.

ACKNOWLEDGMENTS

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